

Detailed Theory: Implementation of Static Routing, RIP, IGRP, EIGRP, and OSPF

1. Static Routing Implementation

Overview

- Static routing involves manually adding routes to a router's routing table.
- Useful for small or stable networks or as backup routes.

Basic Configuration Steps (Cisco IOS):

1. Enter global configuration mode:

shell

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Router> enable Router# configure terminal

2.

Configure a static route:

scss

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Router(config)# ip route [destination_network] [subnet_mask] [next_hop_ip | exit_interface]

Example:

nginx

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```
ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

3.

Verify configuration:

sql

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```
Router# show ip route static
```

4.

Remove static route:

SCSS

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```
Router(config)# no ip route [destination_network] [subnet_mask] [next_hop_ip]
```

Notes:

- Next hop can be an IP address or directly specify the outgoing interface.
- Static routes have administrative distance 1 by default (most trusted).

2. RIP (Routing Information Protocol) Implementation

Overview

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RIP is a distance vector routing protocol using hop count as the metric.

- Version 1 is classful, Version 2 supports classless and VLSM.
- Sends updates every 30 seconds.

Basic Configuration Steps:

1.

Enable RIP routing:

arduino

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```
Router(config)# router rip
```

2.

Specify networks to advertise:

SCSS

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Router(config-router)# network [network_address]

Example:

arduino

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Router(config-router)# network 192.168.1.0 Router(config-router)# network 10.0.0.0

3.

Enable RIP version 2 (optional but recommended):

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Router(config-router)# version 2

4.

Disable auto-summary (important for discontinuous subnets):

scss

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Router(config-router)# no auto-summary

5.

Verify configuration:

arduino

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Router# show ip protocols Router# show ip route rip

Key Points:

- Maximum hop count is 15.
 - Use on small to medium-sized networks.
 - RIP updates are broadcast (v1) or multicast (v2).
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3. IGRP (Interior Gateway Routing Protocol) Implementation

Overview

- Cisco proprietary distance vector protocol.
- Uses composite metrics: bandwidth, delay, load, reliability.
- Superseded by EIGRP.

Basic Configuration Steps:

1. Enable IGRP routing:

SCSS

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```
Router(config)# router igrp [autonomous_system_number]
```

Example:

arduino

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```
Router(config)# router igrp 100
```

2.

Advertise networks:

scss

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Router(config-router)# network [network_address]

Example:

arduino

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Router(config-router)# network 192.168.0.0

3.

Verify configuration:

arduino

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Router# show ip protocols Router# show ip route igrp

Notes:

- IGRP supports up to 255 hop count.
 - Now largely deprecated in favor of EIGRP.
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4. EIGRP (Enhanced Interior Gateway Routing Protocol) Implementation

Overview

- Cisco proprietary hybrid routing protocol.
- Uses DUAL algorithm for fast convergence.

- Supports variable-length subnet masks (VLSM) and classless routing.

Basic Configuration Steps:

1. Enable EIGRP routing:

scss

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```
Router(config)# router eigrp [autonomous_system_number]
```

Example:

arduino

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Router(config)# router eigrp 100

2.

Advertise networks:

scss

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Router(config-router)# network [network_address]

Example:

arduino

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```
Router(config-router)# network 10.0.0.0 Router(config-router)# network 192.168.1.0
```

3.

Disable auto-summary (optional):

scss

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```
Router(config-router)# no auto-summary
```

4.

Verify configuration:

arduino

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Router# show ip protocols Router# show ip route eigrp Router# show ip eigrp neighbors

Advanced Features:

- Supports unequal-cost load balancing.
- Uses metrics based on bandwidth and delay primarily.
- Supports authentication for secure routing updates.

5. OSPF (Open Shortest Path First) Implementation

Overview

- Open standard link-state routing protocol.
- Uses cost metric based on bandwidth.
- Supports hierarchical network design with areas.

Basic Configuration Steps:

1. **Enable OSPF routing:**

scss

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```
Router(config)# router ospf [process_id]
```

Example:

arduino

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Router(config)# router ospf 1

2.

Advertise networks with area assignment:

scss

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Router(config-router)# network [network_address] [wildcard_mask] area [area_id]

Example:

arduino

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```
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0 Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

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The wildcard mask is the inverse of the subnet mask.

3.

Verify configuration:

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```
Router# show ip protocols Router# show ip route ospf Router# show ip ospf neighbor Router# show ip ospf interface
```

Advanced Concepts:

- Backbone Area (Area 0) is mandatory.
- Supports route summarization on ABRs.
- Fast convergence due to link-state updates.

Summary Table

Protocol	Type	Configuration Key Commands	Features	Use Case
Static	Manual	ip route [dest] [mask] [next-hop]	Simple, no overhead	Small or stable networks
RIP	Distance Vector	router rip, network [net], version 2	Hop count metric, max 15 hops	Small to medium networks
IGRP	Distance	router igrp [AS], network [net]	Composite metric, Cisco only	Legacy Cisco networks

	Vector			
EIGRP	Hybrid	router eigrp [AS], network [net]	Fast convergence, VLSM	Medium to large Cisco networks
OSPF	Link-State	router ospf [process], network [net] [wildcard] area [area]	Fast, scalable, hierarchical	Large enterprise networks